

1. CORE INSPECTION METADATA

Timestamp	2026-09-23T22:09:28.600881
Primary Finding	DENT
Severity Layer	HIGH
Detections Logged	1
Analysis Engine	YOLOv11-Nano (Local Edge)

2. ACQUIRED IMAGERY & BOUNDING BOXES



Fig 1. AI-annotated structural imagery overlay.

3. RAW DETECTION COORDINATES

ID	DEFECT CLASS	CONFIDENCE	SEVERITY	BOUNDING BOX [X, Y, W, H]
01	DENT	86.0%	HIGH	[0.430, 0.418, 0.310, 0.207]

4. RAG CONTEXT RETRIEVAL

Query N/A

Source Manual ac_43.13-1b_w-chg1.pdf (Page 150)

Extracted Snippet: *"sustain these stresses, and then allow them to return into the structure. The repair must be equal to the original structure, but not stronger or stiffer, which will cause stress concentrations or alter the resonant frequency of the structure. c. All-metal aircraft are made of very thin sheet metal, and it is possible to restore the strength of the skin without restoring its rigidity. All repairs should be made using the same type and thickness of material that was..."*

5. SYNTHESIZED REPAIR PROTOCOL

Generated via Local Phi-3 Mini strictly grounded in retrieved maintenance documentation.

- 1 Step 1**
Isolate the damaged dented area to prevent further stress and potential damage to surrounding aircraft skin.
- 2 Step 2**
Remove any loose or damaged material from the dented area to prepare for the repair process.
- 3 Step 3**
Apply a suitable material of the same type and thickness as the original aircraft skin to the dented area, ensuring it is compatible with the aircraft's structure.
- 4 Step 4**
Shape the applied material to conform precisely to the original contour of the aircraft skin, using specialized tools to avoid introducing new stresses or altering the resonant frequency.
- 5 Step 5**
Allow the repaired area to cool down and stabilize under controlled conditions, ensuring the repair is sustainable under operational stresses.